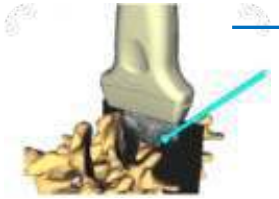
# Pivot calibration



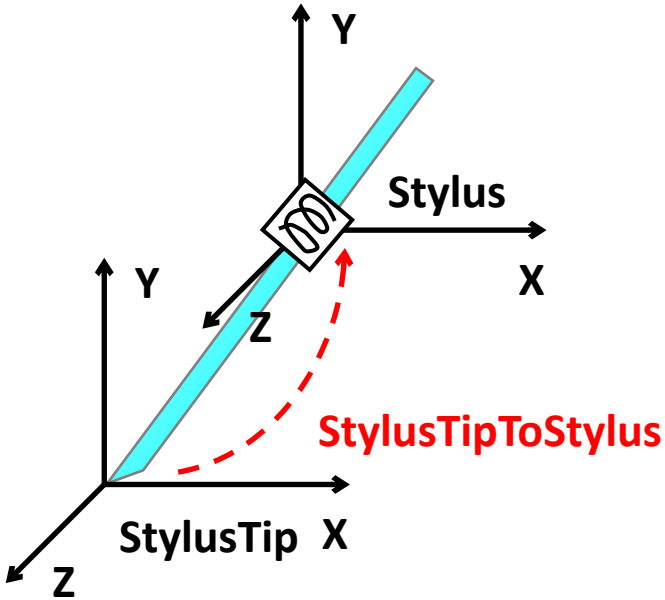
*SlicerIGT Tutorial Series*

# Introduction

- A ***pointer tool*** is used in almost every medical navigation systems. It can be a tracked stylus, needle, or any tracked object that has a pointy part.
- A position marker (e.g. electromagnetic sensor, or optical marker) is used to track the position of the pointer tool.
- But the marker cannot be at the very tip of the pointer. Therefore, we need to determine where is the ***pointer tip*** relative to the pointer marker.
- The names ***Pointer***, ***Stylus***, and ***Needle***. Are used interchangeably in this tutorial.

# Stylus coordinate systems

- There are two coordinate systems associated with a stylus tool:
  - Stylus
    - Origin: center of the position marker
  - StylusTip
    - Origin: tip of the stylus.
- The StylusTipToStylus transformation is computed using the ***Pivot Calibration*** algorithm.




: position marker

# Pivot Calibration in a Real Hardware Setup

- When performing a pivot calibration with a tracked needle
  - Find a secure position to pivot in the Reference coordinate system
  - You must not shift the tip of the needle, only the shaft of the needle may move relative to the tip while you pivot

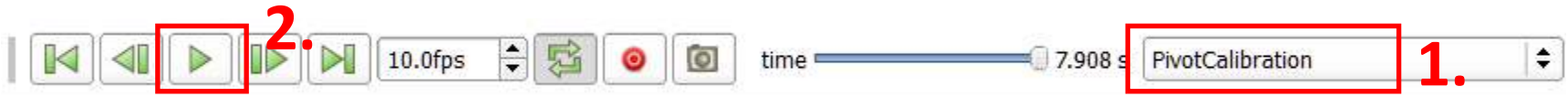
# Example 1.

# Load example data

- Load **SlicerIGT-Data/Skull_StylusCalibration.mrb**
- In **Data** module, right click on **StylusToReference**, *Insert transform*
- Rename new transform to **StylusTipToStylus**
- Create a needle model
- In **Data** module Drag-and-drop needle model on **StylusTipToStylus**
- Make sure the **Sequences** toolbar is visible

# Pivot calibration

- In **IGT / Pivot Calibration** module select *Input*: **StylusToReference**, *Output*: **StylusTipToStylus**
- In the Sequences toolbar, select PivotCalibration and start playback with the play button

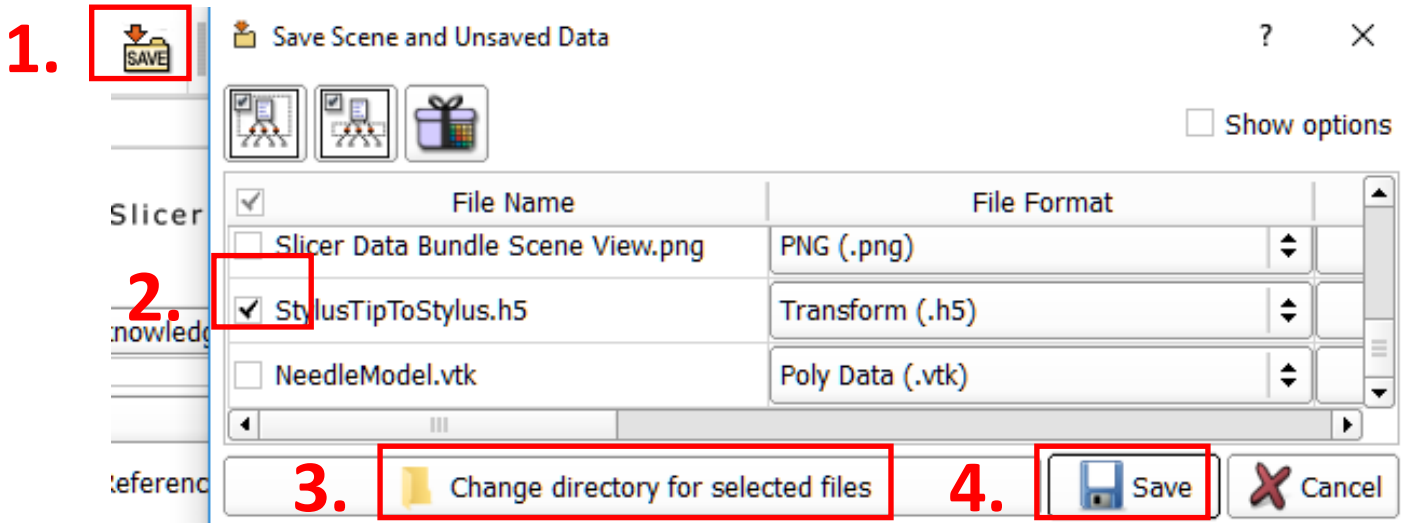



- Press **Start Pivot Calibration** and wait until message at the botton of module widget displays ***Root-mean-square error*** value.
- This value should be under 1 mm

# Pivot calibration

# Spin calibration

- Press **Start Spin Calibration**
- Wait until **Root-mean-square error** value updated
- With this particular stylus design, **StylusTipToStylus** transform does not change significantly
- Save StylusTipToStylus transform into a ***.h5*** file



# Example 2.

(PLUS required. This is method is getting obsolete.)

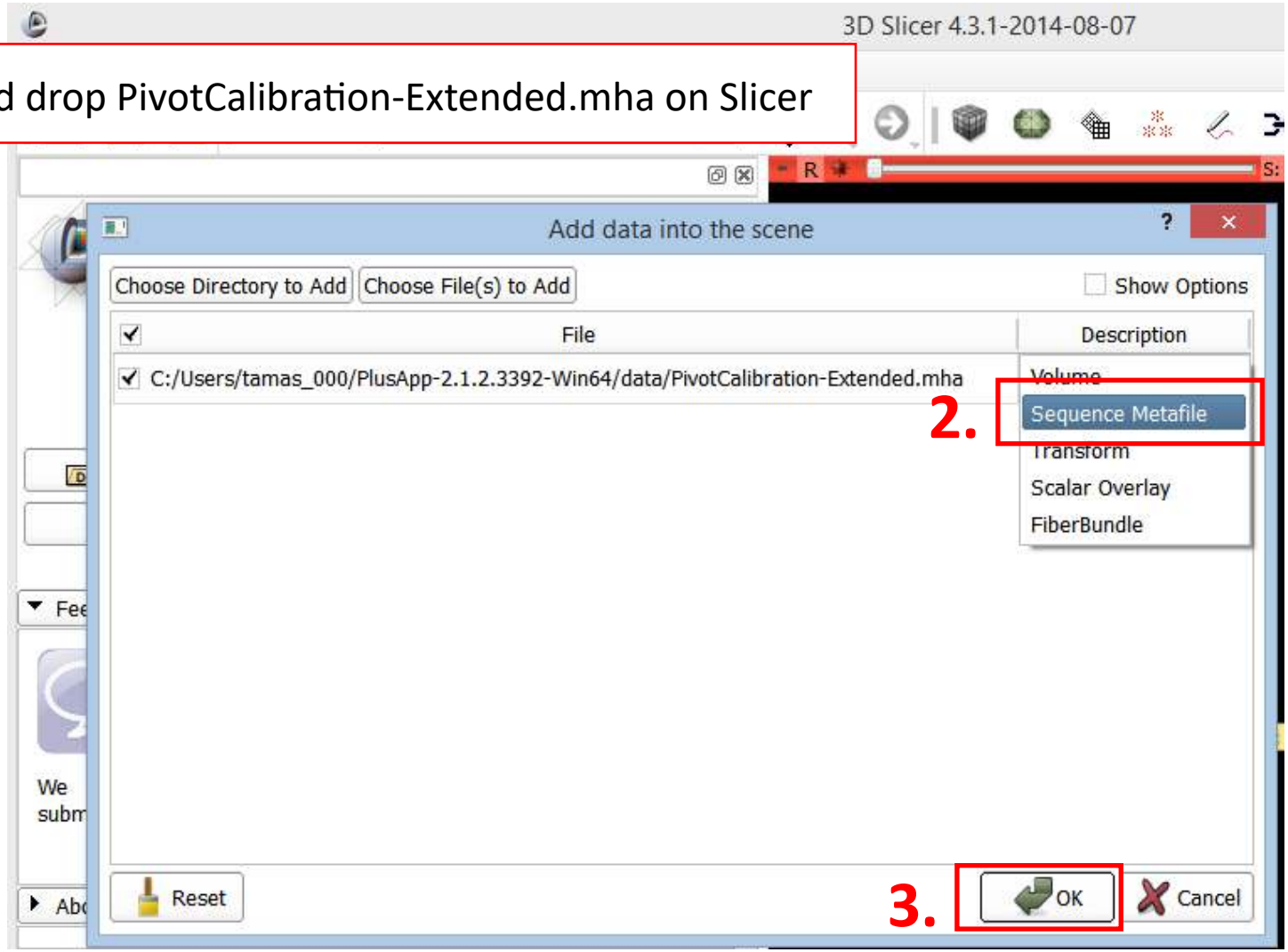
# Pivot calibration in Slicer

- There is a data sequence in SlicerIGT-Data which was recorded using a pivoting stylus: **PivotCalibration.mha**
- The stylus tip was pivoting relative to a tracked **Reference** object. Therefore we need **StylusToReference** transformation to find the **StylusTip** (as the pivot point).
- Run **PivotCalibrationExtend.bat** to get **PivotCalibration-Extended.mha** the /**data** folder of your PLUS installation. E.g. in c: \users\you\PlusApp-2.1.2.3392-Win64\data\

# Load pivot calibration sequence

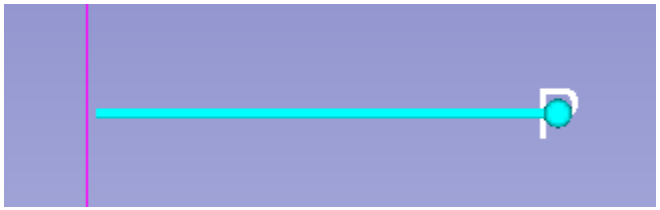
**1.**

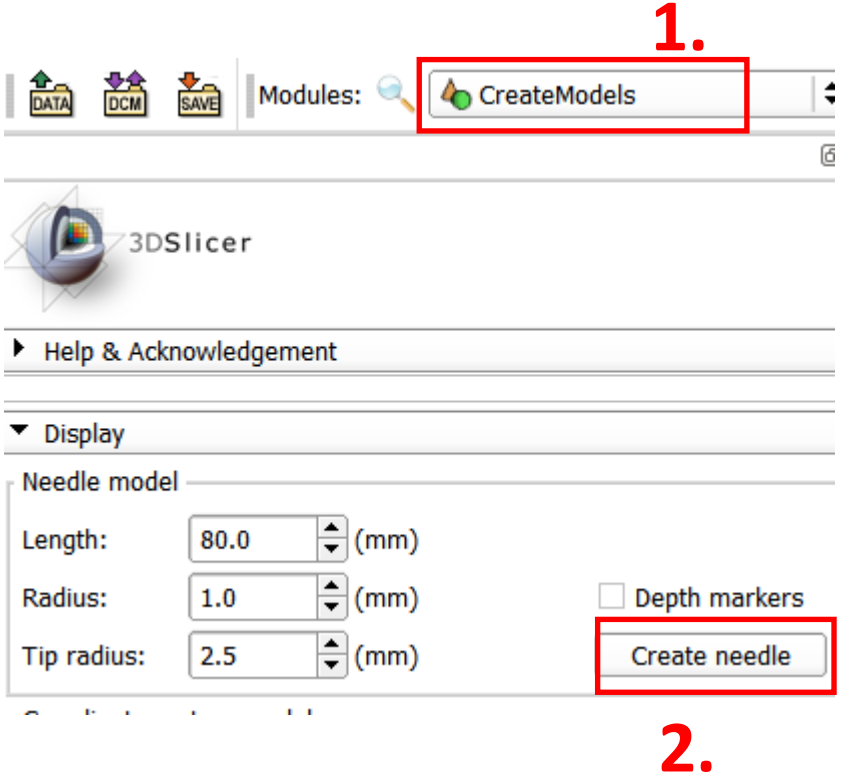
Drag and drop PivotCalibration-Extended.mha on Slicer



# Example: Pivot calibration in Slicer

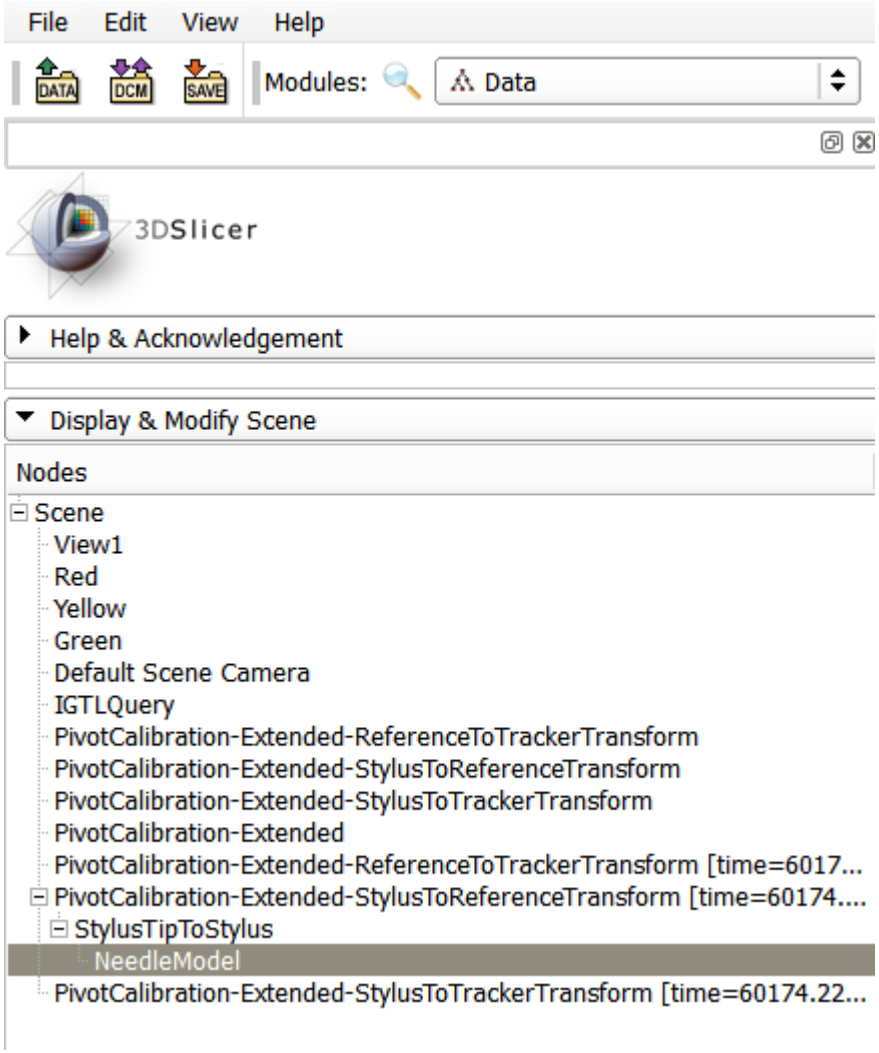
- Create a needle model using **IGT/Create Models** to represent the stylus tool.
- A needle model should appear in the scene that looks like this:



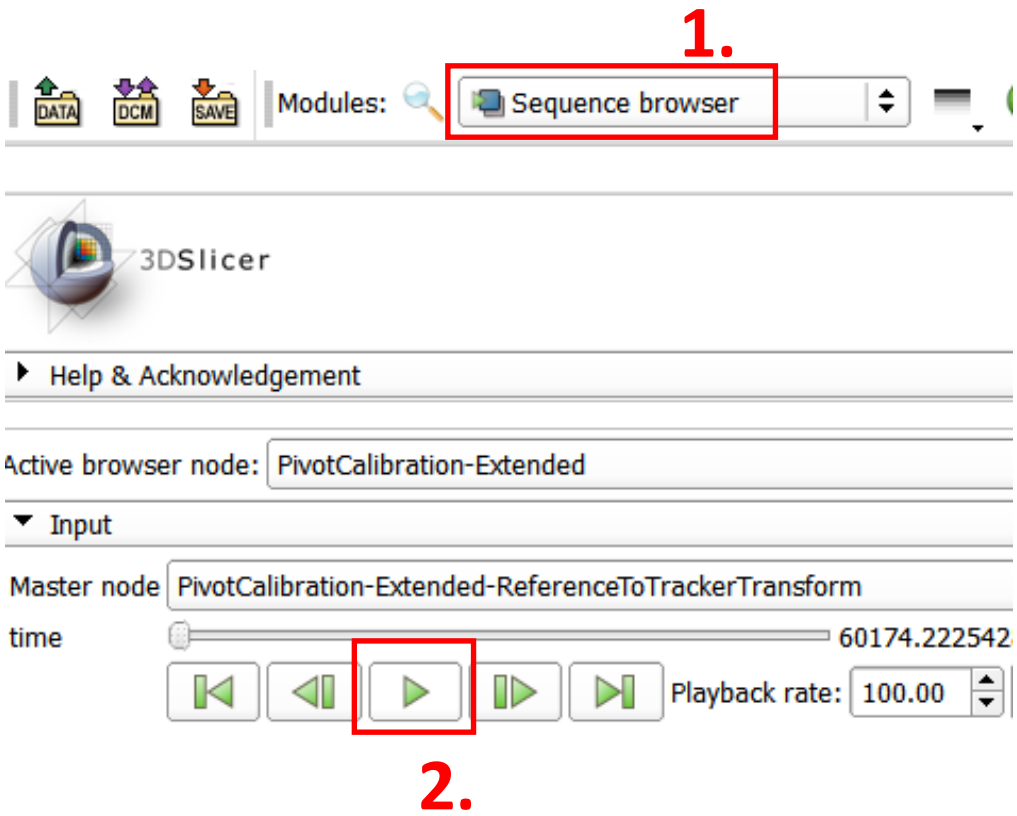


# Example: Pivot calibration in Slicer

- Create the transform hierarchy for the scene:
- Select the ***Transforms*** module and create a new transform, named **StylusTipToStylus**.
- Set up the transform hierarchy in the Data module as seen in the figure.

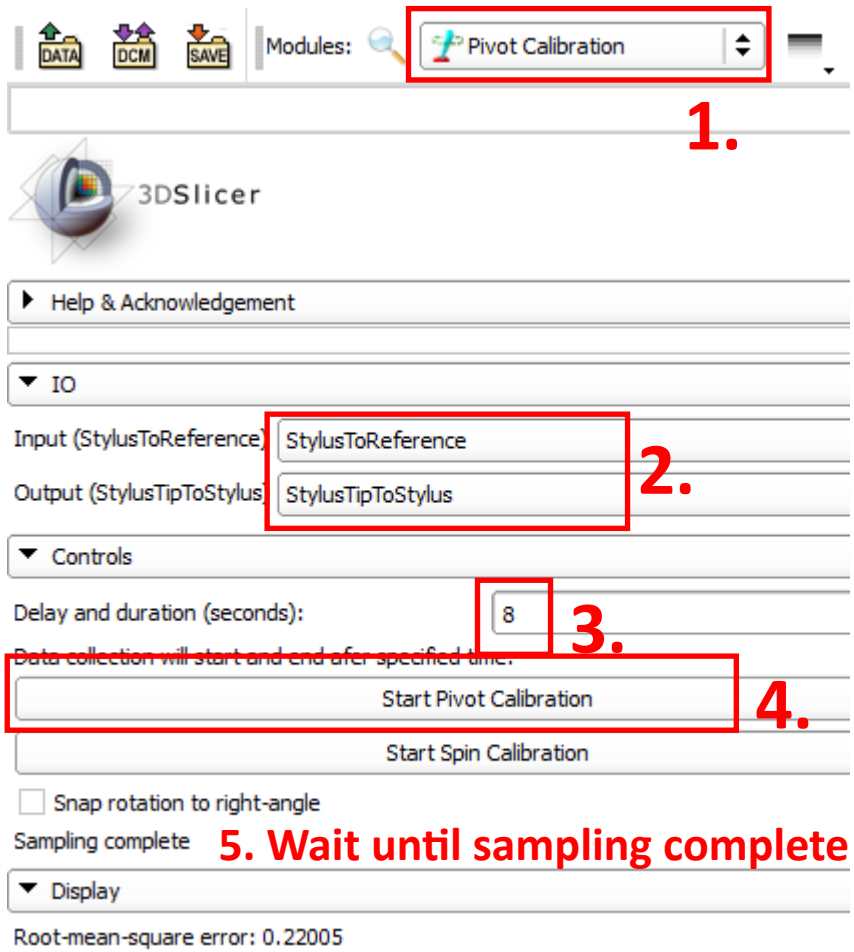


# Play the recorded sequence



# Example: Pivot calibration in Slicer

- Initially, you should see the tip of the stylus swinging.
- Follow the steps shown in red.
- After pivot calibration, the tip of the stylus should be relatively stable (within about 0.5 mm)



# Saving the result

- Save the Slicer scene to save your **StylusTipToStylus** calibration transformation.
- The **.mrml** file in the scene folder contains a human-readable format of the transformation matrix.
- The **StylusTipToStylus.h5** file in the scene folder contains encrypted numbers, but if you drag and drop it on Slicer, it will recover your calibration transformation.